A CONNECTIVITY-SENSITIVE UPPER BOUND FOR LIST-DISTINGUISHING COLORING

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Editorial note. Original author statement, entire paired proof, definitions and supporting arguments retained. No added mathematical hypothesis, proof repair or correctness certification.

Editorial note. Full article context is retained to preserve internal references, prerequisite constructions, and native diagrams. Only this task primary statement is counted.

Editorial note. Mechanical rendering adaptation: renamed only the earlier of three duplicate source labels (405,850,976), preserving the later original label and the original last-definition reference semantics; no mathematics or figure modified.

Allowed standard prerequisites (editorial).

- Standard graph, matroid and enumerative combinatorics. The source article definitions, cited standard results and bibliography are retained.. Original prerequisites; no new assumptions or author-proof replacements.

1. Introduction

All graphs considered in this paper are simple, finite, and connected, and all our results apply only to finite graphs. In [1], Albertson and Collins studied the distinguishing number of a graph, and Collins and Trenk [8] introduced the distinguishing chromatic number of a graph. A coloring $h : V_G \to \{1, ..., r\}$ of the vertices of a graph $G = (V_G, E_G)$ is *r-distinguishing* provided no nontrivial automorphism of G preserves all of the colors of the vertices. The *distinguishing number* of G, denoted by $D(G)$, is the minimum integer r such that G has an r-distinguishing coloring. The coloring h is *r-proper distinguishing* provided h is r-distinguishing and a proper coloring. The *distinguishing chromatic number* of G, denoted by $\chi_D(G)$, is the minimum integer r such that G has an r-proper distinguishing coloring. Collins and Trenk [8] obtained a Brooks' Theorem type upper bound for $\chi_D(G)$.

Theorem 1.1. (Collins and Trenk [8, Theorem 4.5]) *Let G be a connected graph. Then $\chi_D(G) \leq 2\Delta(G) - 1$ unless G is either the complete bipartite graph $K_{\Delta(G),\Delta(G)}$, or the cycle graph C_6. In these cases, $\chi_D(G) = 2\Delta(G)$.*

Recently, Ferrara et al. [11] extended the notion of distinguishing proper coloring to a list distinguishing proper coloring. Given an assignment $L = L(v)_{v \in V_G}$ of lists of available colors to the vertices of G, we say that G is *(properly) L-distinguishable* if there is a (proper) distinguishing coloring f of G such that $f(v) \in L(v)$ for all $v \in V_G$. The *list-distinguishing number* of G, denoted by $D_L(G)$, is the minimum integer k such that G is L-distinguishable for any list assignment L with $|L(v)| = k$ for all $v \in V_G$. Similarly, the *list-distinguishing chromatic number* of G, denoted by $\chi_{D_L}(G)$, is the minimum integer k such that G is properly L-distinguishable for any list assignment L with $|L(v)| = k$ for all $v \in V_G$. If $\chi(G)$ is the chromatic number of G, then the following holds:

$$\chi(G) \leq max\{\chi(G), D(G)\} \leq \chi_D(G) \leq \chi_{D_L}(G).$$

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The second inequality was stated in [8]. Since all lists can be identical, the third inequality holds.

1.1. **Motivation.** Our next example shows that $\chi_{D_L}(G) \neq \chi_D(G)$ in general. This motivates the study of list-distinguishing chromatic numbers. Consider the graph G given in Figure 1. Since G is bipartite and asymmetric, we have $\chi_D(G) = 2$. Consider the following assignment of lists: $L(v_0) = \{1, 2\}$, $L(v_1) = \{1, 3\}$, $L(v_2) = \{2, 3\}$, $L(v_3) = \{1, 4\}$, $L(v_4) = \{2, 4\}$, $L(v_5) = \{3, 4\}$, and for $0 \leq i \leq 6$, $L(s_i)$ can be an arbitrary set of two elements. Then there is no proper coloring of G from $\{L(v)\}_{v \in V_G}$, since the set $\{v_0, \ldots, v_5\}$ does not have a proper coloring from the given lists. Thus, $\chi_{D_L}(G) \neq 2$. Moreover, $\chi_{D_L}(G) = 3$.

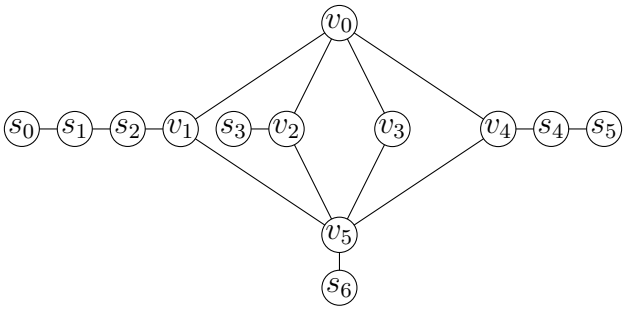


FIGURE 1. *Graph G where $\chi_{D_L}(G) \neq \chi_D(G)$*

1.2. **Structure of the paper.** In Section 3, inspired by Theorem 1.1, the first two authors prove that if G is connected, then $\chi_{D_L}(G) \leq 2\Delta(G) - 1$, and the equality holds if G is $K_{\Delta(G),\Delta(G)}$ or C_6. In view of the graph in Figure 1, this strengthens Theorem 1.1. We remark that $\chi_{D_L}(G) \leq \Delta(G)$ if G is a connected unicyclic graph of girth at least 7 and $\Delta(G) \geq 3$ (see Theorem 3.1, Remark 3.2).

Borodin and Kostochka [5], Catlin [6], and Lawrence [13] independently improved the Brooks-upper bounds for $\chi(G)$ for graphs omitting small cliques by applying a result of Lovász from [14]. In particular, they proved that if $K_r \not\subseteq G$, where $4 \leq r \leq \Delta(G)+1$, then $\chi(G) \leq \frac{r-1}{r}(\Delta(G)+2)$. In Section 4, the first two authors work in a similar fashion to reduce the upper bound of $\chi_{D_L}(G)$ from Theorem 3.1 for certain graphs by applying the same result of Lovász. In particular, we show that if G is $(n - r + 1)$-connected and does not contain a complete bipartite graph $K_{r-1,r-1}$ where $7 \leq r \leq \Delta(G) + 1$, then $\chi_{D_L}(G) \leq 2\Delta(G) - (3\lfloor \frac{(\Delta(G)+1)}{r} \rfloor - 2)$ (see Theorem 4.3).

In Section 5, we prove two upper bounds for $\chi_{D_L}(G)$ in terms of the coloring number $Col(G)$ and the list chromatic number $\chi_L(G)$ (see Theorem 5.1):

(1) $\chi_{D_L}(G) \leq Col(G)D_L(G)$.
(2) If $Aut(G) \cong \Sigma$ where Σ is a finite abelian group so that $Aut(G) \cong \prod_{1 \leq i \leq k} \mathbb{Z}_{p_i^{n_i}}$ for some k, where $p_1, ..., p_k$ are primes not necessarily distinct, then $\chi_{D_L}(G) \leq \chi_L(G) + k$.

In Theorem 5.1, we also give new examples to show that the result mentioned in (2) is sharp. We also determine the list-distinguishing chromatic number of the book graphs.

2. Connected graphs with maximum degree two

Let P_n be a path of n vertices, C_n be a cycle of n vertices, K_n be a complete graph of n vertices, and $K_{n,m}$ be a complete bipartite graph with bipartitions of size n and m.

Fact 2.1. *The following holds:*

(1) If T is a tree, then $\chi_D(T) = \chi_{D_L}(T)$ (see [11, Theorem 7]*).*
(2) If T is a tree, then $\chi_D(T) \leq \Delta(T) + 1$ (see [8, Theorem 3.4]*).*
(3) If G is a graph, then $\chi_D(G) \leq \chi_{D_L}(G)$.

(4) $\chi_D(P_{2t}) = 2$, $\chi_D(P_{2t+1}) = 3$, $\chi_D(C_4) = 4$, $\chi_D(C_5) = 3$, $\chi_D(C_6) = 4$, $\chi_D(C_{2n}) = 3$, $\chi_D(C_{2m+1}) = 3$ *for any* $t \geq 1$, $n \geq 4$, *and* $m \geq 3$ *(see* [8, Theorem 2.2]*).*

Proposition 2.2. $\chi_{D_L}(P_{2t}) = 2$, $\chi_{D_L}(P_{2t+1}) = 3$, *and* $\chi_{D_L}(C_4) = 4$ *for any* $t \geq 1$.

Proof. By Fact 2.1(1,4), we have $\chi_{D_L}(P_{2t}) = \chi_D(P_{2t}) = 2$ and $\chi_{D_L}(P_{2t+1}) = \chi_D(P_{2t+1}) = 3$. Since $4 = \chi_D(C_4) \leq \chi_{D_L}(C_4) \leq 4$, we have $\chi_{D_L}(C_4) = 4$. □

Proposition 2.3. *The following holds:*

(1) If $L = \{L(v)\}_{v \in V_{C_5}}$ *is an assignment of lists of size* 3 *to* V_{C_5}, *and* $|\bigcup_{v \in V_{C_5}} L(v)| \neq 3$, *then* C_5 *is properly* L*-distinguishable.*

(2) If $L = \{L(v)\}_{v \in V_{C_6}}$ *is an assignment of lists of size* 4 *to* V_{C_6}, *and* $|\bigcup_{v \in V_{C_6}} L(v)| \neq 4$, *then* C_6 *is properly* L*-distinguishable.*

Proof. Fix $k \in \{5, 6\}$. Let $v_0, ..., v_{k-1}$ be an enumeration of V_{C_k} in clockwise order. We define a proper distinguishing coloring f of C_k in each case so that $f(v_i) \in L(v_i)$ for all $0 \leq i \leq k-1$.

(1). For each $0 \leq i \leq 2$, pick $c_i \in L(v_i)$ such that c_0, c_1, and c_2 are pairwise distinct. Let, $f(v_i) = c_i$, $f(v_3) \in L(v_3) \setminus \{c_1, c_2\}$, and $f(v_4) \in L(v_4) \setminus \{c_0, f(v_3)\}$.

(2). For each $0 \leq i \leq 3$, pick $c_i \in L(v_i)$ such that c_0, c_1, c_2, and c_3 are pairwise distinct. Let, $f(v_i) = c_i$, $f(v_4) \in L(v_4) \setminus \{c_1, c_2, c_3\}$, and $f(v_5) \in L(v_5) \setminus \{c_0, f(v_4), c_3\}$. □

Proposition 2.4. *Fix* $n \geq 3$. *Let* $L = \{L(v)\}_{v \in V_{C_{2n+1}}}$ *be an assignment of lists of size* 3 *to* $V_{C_{2n+1}}$, *and* $|\bigcup_{v \in V_{C_{2n+1}}} L(v)| \neq 3$, *then* C_{2n+1} *is properly* L*-distinguishable.*

Proof. Since C_{2n+1} is connected, there exist two vertices x and y_1 in C_{2n+1} such that $L(x) \neq L(y_1)$ and $\{x, y_1\} \in E_{C_{2n+1}}$. We color C_{2n+1} as follows:

(*a*) Pick any $c_x \in L(x) \backslash L(y_1)$. Color x with c_x. Let $N(x)$ be the set of vertices adjacent to x.
(*b*) Let $w_1 \in N(x) \backslash \{y_1\}$. Color w_1 with any color c_{w_1} from $L(w_1) \backslash \{c_x\}$.
(*c*) Color y_1 with any color c_{y_1} from $L(y_1) \backslash \{c_{w_1}\}$. We note that $c_x \notin L(y_1)$. Moreover, we can see that c_x, c_{y_1}, and c_{w_1} are pairwise distinct.
(*d*) Let $\{y_2, ..., y_n, w_n, ..., w_2\}$ be the vertices in the path of length $2n-1$ joining y_1 and w_1 in clockwise order (see Figure 2).
(*e*) For any $2 \leq k \leq n$, we color y_k inductively with any color c_{y_k} from $L(y_k) \backslash \{c_x, c_{y_{k-1}}\}$ and color w_k inductively with any color c_{w_k} from $L(w_k) \backslash \{c_x, c_{w_{k-1}}\}$.

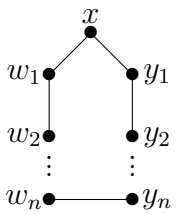


FIGURE 2. *The cycle graph* C_{2n+1}.

CASE 1: If $c_{w_n} \neq c_{y_n}$, then x is the only vertex to get color c_x. Since any nontrivial color-preserving automorphism fixes x, w_1 must map to y_1, but $c_{y_1} \neq c_{w_1}$. Thus, C_{2n+1} is properly L-distinguishable.

CASE 2: If $c_{w_n} = c_{y_n}$, then we recolor w_n with any color d_{w_n} from $L(w_n) \backslash \{c_{y_n}, c_{w_{n-1}}\}$. If $d_{w_n} \neq c_x$, then by the arguments of CASE 1, we are done. If $d_{w_n} = c_x$, then w_n is the only vertex, other than x, to get the color c_x. Fix any nontrivial color-preserving automorphism ϕ. Then ϕ should either fix both x and w_n, or map x to w_n and w_n to x. Similar to CASE 1, ϕ cannot fix x. If ϕ maps x to w_n, then for some $1 \leq m \leq n$, we have either $\{c_{w_m}, c_{y_m}\} = \{c_{w_m}, c_{y_{m+1}}\}$ or $\{c_{w_m}, c_{y_m}\} = \{c_{w_{m-1}}, c_{y_m}\}$, which is a contradiction as $\{y_m, y_{m+1}\}$ and $\{w_m, w_{m-1}\}$ are edges of C_{2n+1}. □

Proposition 2.5. *Fix $n \geq 4$. Let $L = \{L(v)\}_{v \in V_{C_{2n}}}$ be an assignment of lists of size 3 to $V_{C_{2n}}$, and $|\bigcup_{v \in V_{C_{2n}}} L(v)| \neq 3$, then C_{2n} is properly L-distinguishable.*

Proof. Since C_{2n} is connected, assume x, y_1, w_1, c_x, c_{y_1}, and c_{w_1} as in the proof of Proposition 2.4. The following facts will be useful for our proof:

(a) $c_x \notin L(y_1)$,
(b) c_x, c_{y_1}, and c_{w_1} are pairwise distinct.

Let $\{y_2, ..., y_{n-1}, z, w_{n-1}, ..., w_2\}$ be the vertices in the path of length $2n-2$ joining y_1 and w_1 in a clockwise order (see Figure 3). For $2 \leq k \leq n-1$, assume c_{y_k} and c_{w_k} as in the proof of Proposition 2.4. Let $L'(z) = L(z) \setminus \{c_x, c_{y_{n-1}}, c_{w_{n-1}}\}$.

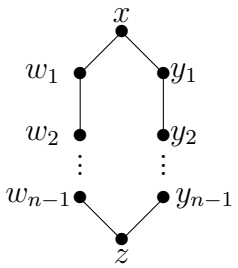


FIGURE 3. *The cycle graph C_{2n}.*

CASE 1: Assume $L'(z) \neq \emptyset$. Then assigning any color from $L'(z)$ to z will result in a proper distinguishing coloring since the only vertex colored by c_x is x and $c_{w_1} \neq c_{y_1}$.

CASE 2: Let $L'(z) = \emptyset$. In order to keep the coloring proper we must assign color c_x to z. Thus, only x and z are colored with c_x. Let f be the coloring of C_{2n} obtained thus far. Without loss of generality, we may assume that there is a nontrivial automorphism ϕ preserving f. Then ϕ should either fix both x and z, or map x to z and z to x. Similar to CASE 1, ϕ cannot fix x. Thus, ϕ maps x to z and z to x.

SUBCASE 2.1: Assume $\phi(y_1) = y_{n-1}$. Then we must have $c_{y_1} = c_{y_{n-1}}$. Define f' as follows:

$$f'(v) = \begin{cases} f(v) & \text{if } v \neq y_1, \\ c & \text{where } c \in L(y_1) \setminus \{c_{y_1}, c_{y_2}\}. \end{cases} \qquad (*)$$

Since $c_x \notin L(y_1)$ by (a), we have that $c \neq c_x$. Thus, f' is a proper coloring. The following claim states that f' is a proper distinguishing coloring.

Claim 2.6. *The only automorphism preserving f' is the identity.*

Proof. Let ψ be a nontrivial automorphism preserving f'. Similar to CASE 1, ψ cannot fix x. Assume that ψ maps x to z and z to x. Since $c \neq c_{y_1} = c_{y_{n-1}}$, we have $\psi(y_1) = w_{n-1}$. Consequently, $\psi(w_1) = y_{n-1}$. Thus, $c_{y_1} = c_{y_{n-1}} = f'(y_{n-1}) = f'(w_1) = f(w_1) = c_{w_1}$, which contradicts (b). □

SUBCASE 2.2: If $\phi(y_1) = w_{n-1}$, then an argument analogous to SUBCASE 2.1 applies. □

Theorem 2.7. *The entries in the following table are correct.*

Graph	$\chi_{D_L}(G)$
$P_{2t}, t \geq 1$	2
$P_{2t+1}, t \geq 1$	3
C_4	4
C_5	3
C_6	4
$C_{2m+1}, m \geq 3$	3
$C_{2n}, n \geq 4$	3

Table 1.

Proof. In each case, we may assume that the lists are not identical; otherwise, we can color the vertices in a manner identical to the proper distinguishing coloring and apply Fact 2.1(4). If the lists are non-identical, then by Propositions 2.2, 2.3, 2.4, 2.5, and Facts 2.1(3,4), we are done. □

3. Brooks' type upper bounds

Imrich et al. [12, Theorem 3] proved that if H is a connected infinite graph with a finite maximum degree $\Delta(H)$, then $\chi_D(H) \leq 2\Delta(H) - 1$. We apply Theorem 2.7 and modify the algorithm of [12, Theorem 3] suitably to prove the following upper bound for $\chi_{D_L}(G)$ if G is a finite graph:

Theorem 3.1. *Let G be a connected finite graph with maximum degree $\Delta(G)$. Then $\chi_{D_L}(G) \leq 2\Delta(G) - 1$, unless G is $K_{\Delta(G),\Delta(G)}$ or C_6. In these cases, $\chi_{D_L}(G) = 2\Delta(G)$.*

Proof. If $\Delta(G) = 2$, then by Theorem 2.7, we have $\chi_{D_L}(G) = 4$ if and only if G is C_4 or C_6. It is easy to see that $\chi_{D_L}(K_{\Delta(G),\Delta(G)}) = 2\Delta(G)$, and $K_{2,2}$ is C_4. Hence, we may assume that $\Delta(G) \geq 3$ and G is not $K_{\Delta(G),\Delta(G)}$. We show that in this case $\chi_{D_L}(G) \leq 2\Delta(G) - 1$. Let $L = \{L(v)\}_{v \in V_G}$ be an assignment of lists with $|L(z)| = 2\Delta(G) - 1$, for all $z \in V_G$. Let $v \in V_G$ be a vertex with degree $\Delta(G)$ and T be a breadth-first search (BFS) spanning tree of G rooted at v. We use the notation $<$ to denote the BFS order. Also, for $z \in V_G$, by $N(z)$ we denote the set of its neighbors in G and $S(z)$ denotes the set of its siblings in T. We will define partial functions $f_x : V_G \to \bigcup_{z \in V_G} L(z)$ for $x \in V_G$ such that the following holds:

(i) if $x < y$, then $f_x \subseteq f_y$,
(ii) if $f = \bigcup_{x \in V_G} f_x$, then $f(v) \in L(v)$ for all $v \in V_G$,
(iii) the domain of f_x includes all vertices till x in the BFS order.

For the base case, we assign pairwise different colors to v and to members of $N(v) = \{v_1, \ldots, v_{\Delta(G)}\}$ from the respective lists, say $c_v \in L(v)$ and $c_{v_i} \in L(v_i)$, where $1 \leq i \leq \Delta(G)$. For the rest, we try to keep v to be the only vertex of G with the property of being colored with c_v, while all its neighbors are colored by $c_{v_1}, \ldots, c_{v_{\Delta(G)}}$. We will refer to this property as $(*)$.

We proceed by coloring the vertices of G in their BFS order inductively, as follows: Let $x \in V_G$ be the $<$-minimal element for which no color has been assigned and $z < x$ be the immediate BFS predecessor of x. We define f_x. Let us write,

$$S_<(x) = \{y \in S(x) : y < x\} \text{ and } N_<(x) = \{y \in N(x) : y < x\}.$$

Then f_z has already assigned colors to each member of $S_<(x) \cup N_<(x)$. Define,

$$A_x = L(x) \setminus \Big(\{c_v\} \cup f_z[S_<(x)] \cup f_z[N_<(x)]\Big).$$

If $A_x \neq \emptyset$, then let $f_x = f_z \cup \{\langle x, c\rangle\}$, where $c \in A_x$. If $A_x = \emptyset$, then since $|N(x)| \leq \Delta(G)$ and $|S(x)| \leq \Delta(G) - 2$, we must have used $|S_<(x)| + |N_<(x)| = 2\Delta(G) - 2$ many colors from $L(x)$ to color the neighbors and siblings that come before x. Thus, the following holds:

(a) $c_v \in L(x)$,

(*b*) $S_<(x) = S(x)$ and $N_<(x) = N(x)$,
(*c*) $|S(x)| = \Delta(G) - 2$ and $|N(x)| = \Delta(G)$,
(*d*) $N(x) \cap S(x) = \emptyset$,
(*e*) for all $y \in S(x) \cup N(x)$, we have $f_z(y) \in L(x)$.

In this case, let $f_x = f_z \cup \{\langle x, c_v\rangle\}$. In either case, f_x has (i) and (iii). Finally, set $f = \bigcup_{x \in V_G} f_x$. It is clear that f satisfies (ii), and whenever $f(x) = c_v$ where $x \neq v$, then $(a) - (e)$ must hold. Moreover, if no vertex other than v has property $(*)$, then it is easy to see that f is a proper distinguishing coloring. However, suppose there is a vertex other than v, say x, with property $(*)$ and is the $<$-minimal such vertex. Then $(a) - (e)$ are satisfied for x. We modify f in such a way that x has no longer $(*)$, while it remains a proper list coloring. By a careful introspection we will analyze the following cases: [1]

Case 1: If there is $w \in S(x)$ such that $|N(w)| \neq |N(x)|$, then color x with $f(w)$, which can be done by (e). Then there is no color-preserving automorphism of G, which fixes v and maps x to w. Meanwhile, by (d), this keeps the coloring proper, and x has no longer the property $(*)$.

Case 2: For every $z \in S(x)$, we have $|N(z)| = |N(x)|$, and there is a $w \in S(x)$ such that $N(w) \neq N(x)$. By (c), fix $y \in N(x)$ such that $y \notin N(w)$. Since the neighbors of x must have occurred before x in the BFS ordering, we have $f(w), f(y) \in L(x)$ by (e). We color x with $f(w)$. The rest follows the arguments of Case 1.

Case 3: For every $w \in S(x)$, we have $N(w) = N(x)$.

Subcase 3.1: There is a vertex $z \in N(x)$ with no sibling or parent colored with c_v. We need to consider two different cases.

Case A: If $c_v \notin L(z)$, then by (c) at most $(\Delta(G) - 1) + (\Delta(G) - 2) = 2\Delta(G) - 3$ number of colors can appear in the list $L(z)$ that are used for coloring $N(z) \cup S(z)$. Hence, there must be $c \in L(z)$ such that $c \neq f(z)$, which also differs from all the colors assigned to $N(z) \cup S(z)$. We color z with c and x with $f(z)$. Then x no longer has $(*)$ (as no member of $N(v)$ has color c), and this modification keeps the coloring proper. Moreover, z does not have the property $(*)$ as z is colored by c, but $c \neq c_v$, since $c \in L(z)$ and $c_v \notin L(z)$.

Case B: If $c_v \in L(z)$, then color x with $f(z)$ and z with c_v, which can be done by (e). Clearly, x does not have $(*)$, and z does not have $(*)$ either by $(b) - (d)$.

Subcase 3.2: Each $y \in N(x)$ has a sibling or parent colored c_v. By (b), each $y \in N(x)$ and its siblings must have come before x in the BFS order. Hence, their parent cannot have $(*)$, unless it is the root v. There must exist a $z \in N(x)$ such that z is not a child of v. Otherwise, G is $K_{\Delta(G),\Delta(G)}$, which contradicts the assumption that G is not $K_{\Delta(G),\Delta(G)}$. Fix such z. Let $B(z)$ denotes the set of colors assigned to the vertices of $N(z) \cup S(z)$. We claim that the set $L'(z) = L(z) \backslash \{f(z) \cup B(z)\}$ is non-empty. Let p be the parent of z in T. Then $N(z) = S(x) \cup \{p\} \cup \{x\}$, $|S(z)| \leq \Delta(G) - 2$, and $|N(z)| = |S(x) \cup \{p\} \cup \{x\}| = \Delta(G)$. The color c_v was used once for x and once among the vertices of $S(z) \cup \{p\}$. Thus, $|B(z)| \leq 2\Delta(G) - 3$. Since $|L(z)| = 2\Delta(G) - 1$, the set $L'(z)$ is non-empty. Pick $c \in L'(z)$. Using (e), we can color x with $f(z)$ and z with c. We note that z does not get property $(*)$. This is because $c \neq c_v$ since $c_v \in B(z)$. Similarly, as in the previous cases, the new coloring remains a proper list coloring, and v is the only $<$-predecessor of x having $(*)$, and all of them are fixed as long as v is fixed.

By iterating this procedure, we can recolor every node with property $(*)$, and hence we obtain a list-distinguishing proper coloring of G. □

Remark 3.2. Alikhani–Soltani [2, Theorem 3.2] proved that $\chi_D(G) \leq \Delta(G) + 1$ if $\Delta(G) \geq 3$, where G is a connected unicyclic graph. Inspired by that result, we prove that if G is a connected

[1] We note that Case 1 and *Case A* of Subcase 3.1 arise when the lists are nonidentical.

unicyclic graph of girth at least 7, then $\chi_{D_L}(G) \leq \Delta(G)$ if $\Delta(G) \geq 3$. First, we observe that, similar to [8, Lemma 3.2] due to Collins and Trenk, one can prove the following lemma:

Lemma 3.3. *Let (T, z) be a rooted tree, and let $L = \{L(v)\}_{v \in V_T}$ be an assignment of lists. A coloring of (T, z) in which each vertex, colored from its lists, that is colored differently from its siblings and from its parent, is a properly L-distinguishing coloring.*

Let C be the unique cycle in G with a set $\{x_i : 1 \leq i \leq t\}$ of vertices. Then, $G \backslash E_C$ is the union of trees T_{x_i}, $1 \leq i \leq t$, where T_{x_i} has only one common vertex x_i with the cycle C. Assign a list assignment $L = \{L(v)\}_{v \in V_G}$ such that $|L(v)| = \Delta(G)$ for all $v \in V_G$.

- By Theorem 2.7, we have $\chi_{D_L}(C) = 3$ since $t \geq 7$.
- Define $L^1(v) := \text{init}_3(L(v))$ for each vertex $v \in V_G$, where $\text{init}_3(L(v))$ denotes the first 3 elements of $L(v)$, and let $L^1 = \{L^1(v)\}_{v \in V_C}$. Since C is properly L^1-distinguishable (say by coloring f^1), we define $f \restriction V_C := f^1$.

Fix $1 \leq i \leq t$. Working much in the same way as in Alikhani–Soltani [2, Theorem 3.2] and by applying Lemma 3.3, we have $\chi_{D_L}(T_{x_i} \backslash \{x_i\}) \leq \Delta(G)$. Since $|L(v)| = \Delta(G)$ for all $v \in V_G$, color the vertices of $T_{x_i} \backslash \{x_i\}$ uniquely from its list (say by f^i). Define $f \restriction V_{T_{x_i} \backslash \{x_i\}} := f^i$.

If ϕ is a nontrivial automorphism preserving f, then all vertices of C are fixed by ϕ, and consequently all vertices in the T_{x_i}'s are fixed. Thus, f is a proper distinguishing coloring, and we conclude that $\chi_{D_L}(G) \leq \Delta(G)$.

4. An application of Lovász's lemma

For a subset $X \subseteq V_G$ of vertices of a graph G, let $G[X]$ denote the subgraph of G induced by X.

Lemma 4.1. *(Lovász; [14]) If $\sum_{i=1}^{t} x_i \geq \Delta(G) + 1 - t$, then there is a partition of V_G into t-sets $V_1, ..., V_t$ such that $\Delta(G[V_i]) \leq x_i$ for all $1 \leq i \leq t$.*

Definition 4.2. A graph is *k-connected* if it has at least $k+1$ vertices and the removal of $k-1$ or fewer vertices leaves a connected graph. We denote $\text{init}_k(L)$ by the first k elements of a list L.

We apply Lemma 4.1 and Theorem 3.1 to prove the following:

Theorem 4.3. *Fix $7 \leq r \leq \Delta(G)+1$. Let G be a $(n-r+1)$-connected graph of order n such that G does not contain a complete bipartite graph $K_{r-1,r-1}$. Then, $\chi_{D_L}(G) \leq 2\Delta(G) - (3\lfloor \frac{(\Delta(G)+1)}{r} \rfloor - 2)$.*

Proof. Let $t = \lfloor \frac{(\Delta(G)+1)}{r} \rfloor$, $h_i = r - 1$ for all $1 \leq i \leq t-1$, and $h_t = \Delta(G) - r(t-1)$. Clearly,

$$t \geq 1 \text{ and } \sum_{i=1}^{t} h_i = \Delta(G) + 1 - t.$$

By Lemma 4.1, there is a partition of V_G into $V_1, ..., V_t$ such that $\Delta(G[V_i]) \leq h_i = r - 1$, for all $1 \leq i \leq t-1$ and $\Delta(G[V_t]) \leq h_t = \Delta(G) - r(t-1)$.

Claim 4.4. $\chi_{D_L}(G) \leq \sum_{i=1}^{t} \chi_{D_L}(G[V_i])$.

Proof. Let $k_i = \chi_{D_L}(G[V_i])$ for each $1 \leq i \leq t$. Assign any list $L(v)$ to each vertex $v \in V_G$ such that $|L(v)| = \sum_{i=1}^{t} k_i$. We define a coloring f for the vertices of G as follows:

(1) Let $L^1(v) := \text{init}_{k_1}(L(v))$ for each vertex $v \in V_G$.

Let $L^1 = \{L^1(v)\}_{v \in V_{G[V_1]}}$. Since $G[V_1]$ is properly L^1-distinguishable (say by coloring f^1), we define $f \restriction V_{G[V_1]} := f^1$.

(2) For any $1 < i \leq t$, let $L^i(v) := \text{init}_{k_i}(L(v) \setminus \sum_{k=1}^{i-1} L^k(v))$ for each vertex $v \in V_G$.

Let $L^i = \{L^i(v)\}_{v \in V_{G[V_i]}}$. Since $G[V_i]$ is properly L^i-distinguishable (say by coloring f^i), we define $f \restriction V_{G[V_i]} := f^i$.

If ϕ is an automorphism of G preserving the coloring f, then the range of $G[V_i]$ with respect to ϕ is $G[V_i]$, for all $1 \leq i \leq t$. Thus, $\phi \restriction G[V_i]$ is an automorphism of $G[V_i]$ preserving f. Since the above defined colorings of $G[V_i]$'s are distinguishing, we have that $\phi \restriction G[V_i]$ is a trivial automorphism for all $1 \leq i \leq t$. Thus, f is a distinguishing coloring of G.

We can see that f is a proper coloring of G as well. Pick any $x, y \in V_G$ such that $\{x, y\} \in E_G$. If $x, y \in V_{G[V_i]}$ for some $1 \leq i \leq t$, then $f(x) \neq f(y)$, as $f \restriction (V_{G[V_i]})$ is a proper coloring. If $x \in V_{G[V_i]}$ and $y \in V_{G[V_j]}$ such that $i \neq j$, then $f(x) \neq f(y)$, as f is defined in a way such that the sets of colors used to color $G[V_i]$ and $G[V_j]$ are different. □

Claim 4.5. *$\chi_{D_L}(G[V_i]) \leq 2r - 3$ for any $1 \leq i \leq t-1$ and $\chi_{D_L}(G[V_t]) \leq 2(\Delta(G) - r(t-1)) - 1$.*

Proof. Fix $1 \leq i \leq t-1$. We will analyze the following cases:

Case 1: Suppose $|V_{G[V_i]}| < r$. As coloring the vertices with distinct colors from their respective lists of size $|V_{G[V_i]}|$ yields a proper distinguishing coloring and $r \geq 7$, we have

$$\chi_{D_L}(G[V_i]) \leq |V_{G[V_i]}| \leq r - 1 \leq 2r - 3.$$

Case 2: Suppose $|V_{G[V_i]}| \geq r$. Since G is $(n - r + 1)$-connected, we have that $G[V_i]$ is connected. We note that $\Delta(G[V_i]) \leq r - 1$.

Subcase 2.1: Let $\Delta(G[V_i]) < r - 1$. Then, by Theorem 3.1, we have

$$\chi_{D_L}(G[V_i]) \leq 2(\Delta(G[V_i])) \leq 2(r-2) = 2r - 4.$$

Subcase 2.2: Let $\Delta(G[V_i]) = r-1$. Since G contains no $K_{r-1,r-1}$ as subgraphs, neither does $G[V_i]$. Thus, $G[V_i]$ cannot be $K_{\Delta(G[V_i]),\Delta(G[V_i])}$. Moreover, $G[V_i]$ cannot be C_6 as $|V_{G[V_i]}| \geq r \geq 7$. Thus, by Theorem 3.1, we have

$$\chi_{D_L}(G[V_i]) \leq |V_{G[V_i]}| \leq r - 1 \leq 2r - 3.$$

Similarly, we can see that $\chi_{D_L}(G[V_t]) \leq 2(\Delta(G) - r(t-1)) - 1$ since $r - 1 \leq \Delta(G) - r(t-1)$. □

By applying Claims 4.4 and 4.5, we get,

$$\begin{aligned}
\chi_{D_L}(G) &\leq \sum_{i=1}^{t} \chi_{D_L}(G[V_i]) \\
&\leq (t-1)(2r-3) + 2(\Delta(G) - r(t-1)) - 1 \\
&\leq 2\Delta(G) - (3t-2).
\end{aligned}$$

□

5. Two upper bounds and special graphs

Erdős and Hajnal [9] introduced the coloring number of a graph G. The least integer k, such that there exists a well-ordering of the vertices of G in which each vertex has fewer than k neighbors that are earlier in the ordering, is defined as the *coloring number* of G, denoted by $Col(G)$. Erdős–Rubin–Taylor [10] and Vizing [15] independently introduced the list-chromatic number of a graph G. Fix an integer k. We say that G is *k-choosable* if for any assignment $L = L(v)_{v \in V_G}$ of lists of available colors to the vertices of G, there is a proper vertex coloring f of G such that $f(v) \in L(v)$ and $|L(v)| = k$ for all $v \in V_G$. The *list chromatic number* of G, denoted by $\chi_L(G)$, is the minimum integer k such that G is k-choosable. The automorphism group of G, denoted by $Aut(G)$, is the group consisting of

automorphisms of G with composition as the operation. The *join* of graphs G_1, G_2, ...,G_n, denoted by $\bigoplus_{1\leq i\leq n} G_i$, has vertex set $\bigcup_{1\leq i\leq n} V_{G_i}$ and edge set $\bigcup_{1\leq i\leq n} E_{G_i} \cup \{xy : x \in V_{G_i}, y \in V_{G_j}, i \neq j\}$.

Theorem 5.1. *The following hold:*

(1) For any graph $G = (V_G, E_G)$, we have $\chi_{D_L}(G) \leq Col(G)D_L(G)$.
(2) There exists an asymmetric graph G such that $\chi_{D_L}(G) = Col(G)D_L(G)$.
(3) If $Aut(G) \cong \Sigma$, where Σ is an abelian group of order p^m, where p is a prime, then $\chi_{D_L}(G) \leq \chi_L(G) + 1$. Moreover, the bound $\chi_{D_L}(G) \leq \chi_L(G) + 1$ is sharp.
(4) If $Aut(G) \cong \Sigma$, where Σ is a finite abelian group so that $Aut(G) \cong \prod_{1\leq i\leq k} \mathbb{Z}_{p_i^{n_i}}$ for some k, where $p_1, ..., p_k$ are primes not necessarily distinct, then $\chi_{D_L}(G) \leq \chi_L(G) + k$. Moreover, the bound $\chi_{D_L}(G) \leq \chi_L(G) + k$ is sharp.

Proof. (1). Let $c = Col(G)$, $d = D_L(G)$, and $l = \{l(v)\}_{v\in V_G}$ be an assignment of lists with $|l(v)| = cd$, for all $v \in V_G$. Let $\{v_1, ..., v_n\}$ be an enumeration of V_G. We define an assignment $\{l'(v)\}_{v\in V_G}$ of disjoint lists inductively such that $|l'(v)| = d$ for all $v \in V_G$.

Let $l'(v_1)$ be an arbitrary subset of d elements of $l(v_1)$. Fix $i \geq 2$. Suppose $l'(v_1), ..., l'(v_{i-1})$ are defined. We define $l'(v_i)$. Let $v_{k_1}, ..., v_{k_r}$ be the r -vertices of $v_1, ..., v_{i-1}$ that are connected to v_i by an edge. Let

$$X(v_i) = l(v_i) \setminus \bigcup_{1\leq j\leq r} l'(v_{k_j}).$$

Since $c > r$, we have $|X(v_i)| \geq dc - dr = d(c - r) \geq d$. Let $l'(v_i)$ be a d-element subset of $X(v_i)$.

Since $d = D_L(G)$, there is a distinguishing coloring f of G such that $f(v) \in l'(v)$ for all $v \in V_G$. By the definition of $\{l'(v)\}_{v\in V_G}$, f is a proper coloring as well.

(2). Consider the graph G in Figure 1, where $Col(G) = 3$, $D_L(G) = 1$, and $\chi_{D_L}(G) = 3$.

(3). We slightly modify the methods of Collins, Hovey, and Trenk [7, Theorem 4.2]. Let $l = \{l(v)\}_{v\in V_G}$ be an assignment of lists of size $\chi_L(G) + 1$. Define $l'(v) := init_{\chi_L(G)}(l(v))$ and $l' = \{l'(v)\}_{v\in V_G}$. Let f be a proper coloring of G where $f(v) \in l'(v)$ for all $v \in V_G$. Now $\Sigma \cong \mathbb{Z}_{p^m}$. If σ is a generator of $Aut(G)$, then $\tau = \sigma^{p^{m-1}}$ is a nontrivial automorphism of G. Let ω be a nontrivial automorphism of G of order p^t for $t \geq 2$. Clearly, τ is a power of ω. Since τ is a nontrivial automorphism, there is a vertex v not fixed by τ. Then v is not fixed by ω as well. We recolor v by the unique new color from $l(v)\setminus l'(v)$. This new coloring is an l-proper distinguishing coloring of G.

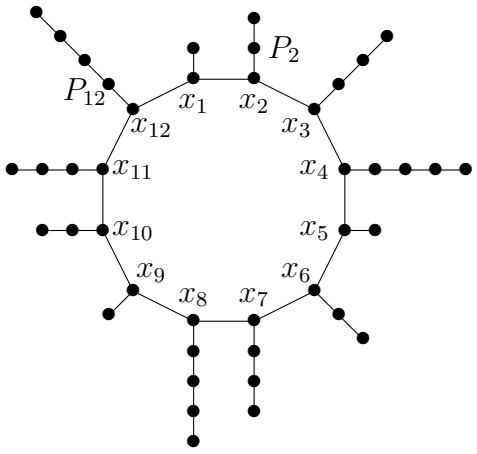


FIGURE 4. *The graph C'_{12}, where $Aut(C'_{12}) = \mathbb{Z}_3$.*

For the second assertion, consider the cycle graph $C_{4n} = (V_{C_{4n}}, E_{C_{4n}})$ for any integer $n \geq 2$. Let $x_1, ..., x_{4n}$ be the vertices of $V_{C_{4n}}$ in a clockwise order. We construct a graph C'_{4n} by adding to each

x_k a path P_k of length r, where $r \in \{1,2,3,4\}$ and $k \equiv r \pmod 4$ (see Figure 4). Now, $Aut(C_{4n})$ is isomorphic to the dihedral group of order $8n$. The added paths in C'_{4n} break all the reflectional symmetries of C_{4n} since any 3 consecutive nodes in C_{4n} have different added paths. However, the added paths of C'_{4n} do not break all of the symmetries of C_{4n} due to its periodicity. Since any automorphism of C'_{4n} can only send a node $x \in V_{C_{4n}}$ to a node $y \in V_{C_{4n}}$ such that P_x and P_y are of the same length, C'_{4n} keeps n out of the $4n$ rotational symmetries of C_{4n}. Thus, $Aut(C'_{4n}) \cong \mathbb{Z}_n$.

Claim 5.2. *$\chi_L(C'_{4n}) = 2$ and $\chi_{D_L}(C'_{4n}) = 3$.*

Proof. Let $H = C'_{4n}$ and $l = \{l(v)\}_{v \in V_H}$ be an assignment of lists of size 2. Let f be a proper coloring of C_{4n} such that $f(v) \in l(v)$ for all $v \in V_{C_{4n}}$.[2] We extend f to a proper coloring of H in a way such that the adjacent vertices of the P_i's get different colors from their respective lists. So, $\chi_L(H) = 2$. We show $\chi_{D_L}(H) = 3$. Any 2-proper coloring f of H must color alternately each of the vertices in both C_{4n} and the added paths. So, there is a nontrivial f-preserving automorphism of H that maps a node $x \in V_{C_{4n}}$ to a node $y \in V_{C_{4n}}$ so that P_x and P_y have the same length. Thus, $\chi_{D_L}(H) \geq \chi_D(H) > 2$. We prove $\chi_{D_L}(H) \leq 3$. Let $l = \{l(v)\}_{v \in V_H}$ be an assignment of lists of size 3. By Theorem 2.7, C_{4n} is properly l'-distinguishable if $l' = \{l(v)\}_{v \in V_{C_{4n}}}$. We extend f to a proper distinguishing coloring of H such that the adjacent vertices of P_i's get different colors from their respective lists. □

(4). The bound $\chi_{D_L}(G) \leq \chi_L(G) + k$ follows from the arguments of [7, Theorem 4.4] and the proof of the first assertion of (3). For the second assertion, we prove that given a finite abelian group $\Gamma = \prod_{1 \leq i \leq k} \mathbb{Z}_{p_i^{n_i}}$ where $n_1, ..., n_k$ are different positive integers, there exists a graph H such that $Aut(H) = \Gamma$ and $\chi_{D_L}(H) = \chi_L(H) + k$. Let $H_i = C'^{n_i}_{4p_i}$ for all $1 \leq i \leq k$ where $C'^{n_i}_{4p_i}$ is the graph constructed in (3). Let $H = \bigoplus_{1 \leq i \leq k} H_i$. Fix any $1 \leq i \leq k$. By Claim 5.2, we have $\chi_L(H_i) = 2$ and $\chi_{D_L}(H_i) = 3$. We note that H_i is triangle free and is not a complete bipartite graph. Moreover, $H_i \ncong H_j$ if $i \neq j$. We recall the following lemma.

Lemma 5.3. (Collins, Hovey, and Trenk [7, Lemma 5.3]) *Suppose each of the graphs $G_1, ..., G_n$, is triangle free, and is not a complete bipartite graph, and also suppose $G_i \ncong G_j$ whenever $i \neq j$. Then, $Aut(\bigoplus_{1 \leq i \leq n} G_i) = \prod_{i=1}^n Aut(G_i)$.*

By applying the techniques in [7, Corollary 5.5] due to Collins, Hovey, and Trenk and the arguments of Claim 4.4, we can prove the following lemma:

Lemma 5.4. *If $Aut(\bigoplus_{1 \leq i \leq n} G_i) = \prod_{i=1}^n Aut(G_i)$, then $\chi_{D_L}(\bigoplus_{1 \leq i \leq n} G_i) = \sum_{i=1}^n \chi_{D_L}(G_i)$.*

By Lemma 5.3, $Aut(H) = \prod_{i=1}^k \mathbb{Z}_{p_i^{n_i}}$. We apply Lemma 5.4 to the graph H to conclude that

$$\chi_{D_L}(H) = \sum_{i=1}^{k} \chi_{D_L}(H_i) = 3k, \text{ and } \chi_L(H) = \sum_{i=1}^{k} \chi_L(H_i) = 2k.$$

□

Question 5.5. *Does there exist a graph G such that $\chi_{D_L}(G) = Col(G)D_L(G)$ and G is not asymmetric?*

5.1. **Book graphs and Friendship graphs.**

Definition 5.6. The *n-book graph* B_n ($n \geq 2$) (see Figure 5) is defined as the Cartesian product of the star graph $K_{1,n}$ and the path graph P_2. We call every C_4 in the book graph B_n a page of B_n. The n pages of B_n are denoted by $v_0w_0v_iw_i$ for $1 \leq i \leq n$. The *friendship graph* F_n ($n \geq 2$) is obtained by joining n copies of the cycle graph C_3 with a common vertex w (see Figure 6).

[2] as if C is an even cycle then the list chromatic number $\chi_L(C)$ of C is 2.

Alikhani and Soltani [4, 3] proved that $D_L(F_n) = D(F_n) = \lceil \frac{1+\sqrt{8n+1}}{2} \rceil$ and $D_L(B_n) = D(B_n) = \lceil \sqrt{n} \rceil$ for any $n \geq 2$. Define

- $A = \{n : (\lceil \sqrt{n} \rceil)^2 - (\lceil \sqrt{n} \rceil) + 2 \leq n \leq (\lceil \sqrt{n} \rceil)^2\}$,
- $B = \{n : (\lceil \sqrt{n} \rceil - 1)^2 < n \leq (\lceil \sqrt{n} \rceil)^2 - (\lceil \sqrt{n} \rceil) + 1\}$.

The purpose of this section is to illustrate the fact that the values of $\chi_{D_L}(B_n)$ and $\chi_D(B_n)$ depend on whether $n \in A$ or $n \in B$, while the values of $D_L(B_n)$ and $D(B_n)$ have a unique form in each case, independently of the value of n (see [4, 3]). We prove that the entries in the following table are correct (see Theorem 5.12 and Remark 5.13).

Graph (G)	$\chi_D(G)$	$\chi_{D_L}(G)$
Book graph (B_n) if $n \in A$	$2 + \lceil \sqrt{n} \rceil$	$2 + \lceil \sqrt{n} \rceil$
Book graph (B_n) if $n \in B$	$1 + \lceil \sqrt{n} \rceil$	$1 + \lceil \sqrt{n} \rceil$
Friendship graph (F_n)	$1 + \lceil \frac{1+\sqrt{8n+1}}{2} \rceil$	$1 + \lceil \frac{1+\sqrt{8n+1}}{2} \rceil$

Table 2.

Observation 5.7. For $n \geq 2$, let f be a coloring of the set of vertices of B_n. Then f is a proper distinguishing coloring of B_n if and only if the following holds:

(*a*) $(f(v_i), f(w_i)) \neq (f(v_j), f(w_j))$ for $0 < i \neq j \leq n$,
(*b*) $f(v_i) \neq f(w_i)$ for $i \leq n$,
(*c*) $f(v_i) \neq f(v_0)$ and $f(w_i) \neq f(w_0)$ for $0 < i \leq n$.

Condition (*a*) makes f distinguishing, whereas conditions (*b*) and (*c*) make it proper.

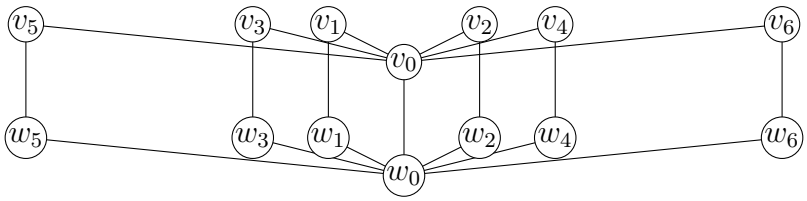


FIGURE 5. *Book graph* B_6.

Lemma 5.8. *For all* $n \geq 2$, $\chi_D(B_n) \leq \lceil \sqrt{n} \rceil + 2$.

Proof. Consider a $\lceil \sqrt{n} \rceil$-distinguishing coloring f of B_n. Using f, we define a $(\lceil \sqrt{n} \rceil + 2)$-proper distinguishing coloring of B_n. Let $C = \{1, \ldots, \lceil \sqrt{n} \rceil + 2\}$ be a color set. Since f is distinguishing, for each pair (v_i, w_i) and (v_j, w_j) with $0 < i \neq j$, we must have $(f(v_i), f(w_i)) \neq (f(v_j), f(w_j))$. Define $f' : B_n \to C$ such that $f'(v_0) = \lceil \sqrt{n} \rceil + 1$, $f'(w_0) = \lceil \sqrt{n} \rceil + 2$, and

$$f'(v_i) = \begin{cases} \lceil \sqrt{n} \rceil + 2 & \text{if } f(v_i) = f(w_i), \\ f(v_i) & \text{otherwise.} \end{cases}$$

Clearly, f' is a $(\lceil \sqrt{n} \rceil + 2)$-proper distinguishing coloring. □

Lemma 5.9. *For all* $n \geq 2$, $\lceil \sqrt{n} \rceil < \chi_D(B_n)$.

Proof. Since $n \geq 2$, we may assume that $\lceil \sqrt{n} \rceil = m + 1$, for some $0 \neq m \in \mathbb{N}$. Suppose we have a proper distinguishing coloring f using only $m + 1$ many colors with color set $C = \{1, \ldots, m + 1\}$. Then conditions $(a) - (c)$ must hold for f from Observation 5.7. The total number of possibilities to color (v_i, w_i) is $(m + 1)^2$ for any $0 < i \leq n$. But f cannot use those pairs $(a, b) \in C \times C$ for which either $a = b$ or $a = f(v_0)$ or $b = f(w_0)$. Thus, the number of color pairs $(a, b) \in C \times C$

that are allowed to color (v_i, w_i) is $(m+1)^2 - 3(m+1) + 3$ (i.e., identical pairs, pairs with the first coordinate $f(v_0)$, and pairs with the second coordinate $f(w_0)$ have to be avoided). But then

$$(m+1)^2 - 3(m+1) + 3 = m^2 + 1 - m \leq m^2 < n.$$

Consequently, (a) in Observation 5.7 cannot be satisfied, i.e., there will be pairs (v_i, w_i) and (v_j, w_j) such that $(f(v_i), f(w_i)) = (f(v_j), f(w_j))$. □

Lemma 5.10. *Let B_n be such that $\lceil \sqrt{n} \rceil = m+1$, for some $0 \neq m \in \mathbb{N}$. Then $\chi_D(B_n) < \lceil \sqrt{n} \rceil + 2$ if and only if $m^2 < n \leq m^2 + m + 1$.*

Proof. Clearly, $m^2 < n \leq (m+1)^2$. Let $C = \{1, \ldots, m+2\}$ be a color set. Fix $a, b \in C$. Color v_0 with a and w_0 with b. The total number of possibilities to color (v_i, w_i) is $(m+2)^2$ for $0 < i \leq n$, among which $m^2 + m + 1$ possibilities are allowed (by removing identical pairs, pairs with the first coordinate a and pairs with the second coordinate b). If $n \leq m^2 + m + 1$, then we can find a coloring f satisfying $(a) - (c)$ such that $f(v_0) = a$ and $f(w_0) = b$. If $n > m^2 + m + 1$, then following the arguments of Lemma 5.9, we cannot find a proper distinguishing coloring. □

Theorem 5.11. *For all $n \geq 2$, we have $\chi_D(B_n) = \lceil \sqrt{n} \rceil + 1$ or $\chi_D(B_n) = \lceil \sqrt{n} \rceil + 2$. Moreover, if $\lceil \sqrt{n} \rceil = m + 1$, for some $0 \neq m \in \mathbb{N}$, then*

(i) $\chi_D(B_n) = \lceil \sqrt{n} \rceil + 1$, whenever $m^2 < n \leq m^2 + m + 1$,
(ii) $\chi_D(B_n) = \lceil \sqrt{n} \rceil + 2$, whenever $m^2 + m + 2 \leq n \leq (m+1)^2$.

Proof. This follows from Lemmas 5.8, 5.9, and 5.10. □

Theorem 5.12. *For all $n \geq 2$, $\chi_{D_L}(B_n) = \chi_D(B_n)$.*

Proof. It is enough to show that $\chi_{D_L}(B_n) \leq \chi_D(B_n)$. Let $\lceil \sqrt{n} \rceil = m+1$, for some $0 \neq m \in \mathbb{N}$. Let $\{L(x_i)\}_{x_i \in V_{B_n}}$ be an assignment of lists with $|L(x_i)| = \chi_D(B_n)$, for every $x_i \in B_n$. Fix $c_{v_0} \in L(v_0)$ and $c_{w_0} \in L(w_0)$ so that $c_{v_0} \neq c_{w_0}$. Fix a pair (v_i, w_i), where $0 < i \leq n$. By $(L(v_i), L(w_i))$, we denote the number of color pairs $(a_i, b_i) \in L(v_i) \times L(w_i)$ such that $a_i \neq c_{v_0}$, $b_i \neq c_{w_0}$, and $a_i \neq b_i$.

Case 1: Assume $\chi_D(B_n) = \lceil \sqrt{n} \rceil + 2$. By Theorem 5.11(ii), we have $n \leq (m+1)^2$. Thus, $|(L(v_i), L(w_i))| \geq m^2 + 3m + 6 \geq (m+1)^2 \geq n$.

Case 2: Assume $\chi_D(B_n) = \lceil \sqrt{n} \rceil + 1$. Since by Theorem 5.11(i), we have $n \leq m^2 + m + 1$, thus $|(L(v_i), L(w_i))| \geq m^2 + m + 1 \geq n$.

In both cases, there is a proper distinguishing coloring f where $f(v_0) = c_{v_0}$ and $f(w_0) = c_{w_0}$. □

Remark 5.13. We remark that $\chi_{D_L}(F_n) = \chi_D(F_n) = 1 + \lceil \frac{1+\sqrt{8n+1}}{2} \rceil$ for every $n \geq 2$. First, we prove that $\chi_D(F_n) \geq 1 + D(F_n)$. If L is a proper distinguishing coloring for F_n, and the color of the two vertices on the base of the i-th triangle is x_i, y_i (see Figure 6), then the following holds:

(1) For every $i \in \{1, ..., n\}$, $x_i \neq y_i$.
(2) For every $i, j \in \{1, ..., n\}$ where $i \neq j$, $\{x_i, y_i\} \neq \{x_j, y_j\}$.
(3) The color of the central vertex w, say z, cannot be x_i or y_i for any $i \in \{1, ..., n\}$.

Thus, $\chi_{D_L}(F_n) \geq \chi_D(F_n) \geq min\{s : \binom{s}{2} \geq n\} + 1 = \lceil \frac{1+\sqrt{8n+1}}{2} \rceil + 1 = D(F_n) + 1$.[3] We show that $\chi_{D_L}(F_n) \leq D(F_n)+1$. Let $L = \{L(v)\}_{v \in V_{F_n}}$ be a list assignment to F_n such that $|L(v)| = D(F_n)+1$. Pick a color for the central vertex w, say $c_w \in L(w)$. Then $L(i) = L(v_i) \backslash \{c_w\}$ has cardinality $D(F_n)$ or $D(F_n) + 1$. By the methods of [4], we can color v_i's and w_i's in a distinguishing way. Moreover, the coloring is proper as well. Consequently, $\chi_{D_L}(F_n) = D(F_n) + 1$.

[3] In [3], Alikhani and Soltani proved that $D(F_n) = min\{s : \binom{s}{2} \geq n\} = \lceil \frac{1+\sqrt{8n+1}}{2} \rceil$.

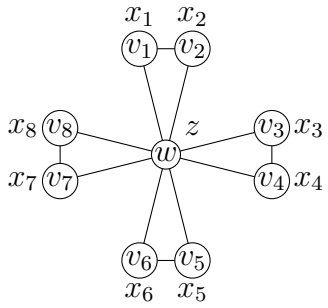


FIGURE 6. *Coloring (L) of the Friendship graph F_4 with central vertex w.*

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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